

Sohan Sharad Pillai

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Education

University of Warwick, BEng in Computer Systems Engineering Sep 2025 – Jun 2028

- **Grade:** Predicted First
- **Relevant Coursework:** Sets and Proofs, Computer Organization and Architecture, Programming for Computer Scientists, Engineering Mathematics, Design of Information Structures, Systems Modelling, Simulation and Computation

The King's School, Kollam, Cambridge International A-Levels Mar 2023 – Jun 2025

- **Grade:** A, A, A
- **Subjects:** Physics, Chemistry, Mathematics

Projects

RISC-V 32-bit CPU Architecture Jun 2026 – Present

- Designing a custom RISC-V ISA physical processor core in Verilog for synthesis on a Lattice iCE40 FPGA (iCEBreaker FPGA V1.1a board) using open-source toolchains.
- Implementing a multi-stage instruction pipeline, Arithmetic Logic Unit (ALU), and custom UART interfaces for low-level hardware control, alongside a rudimentary VGA display controller.
- Tools Used: Verilog, Lattice iCE40, iCEBreaker, Yosys, nextpnr, Assembly

GBA Rom Hacking with Assembly 2023

- Reverse-engineered Game Boy Advance binaries to implement custom game logic. Wrote optimized ARM Assembly and some C routines for the GBA hardware constraints, managing manual memory allocation and instruction cycle timing.
- Tools Used: C, GBA ASM

Game Companion Discord Bot 2020 – 2024

- Architected a high-throughput Discord bot serving 50,000+ users across 1,000 servers. Designed a scalable backend with Python and SQLite to handle concurrent data ingestion of 1M+ records, optimizing query latency for real-time game stat retrieval via packet interception.
- Tools Used: Python, Wireshark, SQLite

AI-Powered A-Level Exam Search Engine 2024

- Prototyped a search engine to sort and query past exam paper questions from 2015 to 2024 using Llama and Elasticsearch, for easy access to answers for enabling more productive revision.
- Tools Used: Llama, Elasticsearch, Python, Javascript, React

Experience

Application Developer Intern, Xploro Technologies – New Delhi, India Apr 2023 – May 2023

- Engineered a cross-platform crypto portfolio application using React Native, implementing real-time data fetching at scalable quantities.
- Collaborated with senior engineers to write production-level code while upholding industry standards.

Data Analyst Intern, Hie Capital Technologies – Mumbai, Maharashtra, India Aug 2024 – Sep 2024

- Developed automated data processing pipelines for large-scale oceanographic datasets using Python, reducing manual data input and analysis by 60%.
- Visualized complex oceanographic datasets, translating telemetry into actionable client reports.

Research & Publications

Genetic Algorithms and Their Effectiveness in NP-Complete Problems: Sudoku Dec 2024

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10.5281/zenodo.17518227

- Implemented Genetic Algorithms in Python to solve NP-Complete Sudoku grids, benchmarking convergence rates against other common algorithms

Technologies

Languages: C++, C, Java, SQL, JavaScript, Python, Rust, Assembly, Bash, Verilog

Technologies: React, SQLite, Pandas, NumPy, OpenCV, NodeJS, QT, Linux, Git, Wireshark, React Native, Flutter, Electron, Yosys, nextpnr